Sample Solutions Jan 2023: Recursion, Differentiation & Integration Questions

Question 1

c) Recursion

$t(1) = 2$

$t(n) = t(n-1) + 3 \quad (n > 1)$

$t(2) = t(2-1) + 3$ — Replace n with 2

$= t(1) + 3$ — $t(1)$ given in the defined sequence

$= 2 + 3$

$t(2) = \boxed{5}$

$t(3) = t(3-1) + 3$ — Replace n with 3

$= t(2) + 3$ — we know $t(2)$ from previous step = 5

$= 5 + 3$

$t(3) = \boxed{8}$

$t(4) = t(4-1) + 3$ — Replace n with 4

$= t(3) + 3$ — we know $t(3)$ from previous step = 8

$= 8 + 3$

$t(4) = \boxed{11}$

g) First Principles — we did not cover

→over

indefinite Integrals

h) i) $\int (4x^3 + 3x^2 - 2x + 7).dx$ $\int (x^n).dx = \frac{x^{n+1}}{n+1} + C$

$= \frac{4x^{3+1}}{3+1} + \frac{3x^{2+1}}{2+1} - \frac{2x^{1+1}}{1+1} + 7x + C$

$= \frac{4x^4}{4} + \frac{3x^3}{3} + \frac{2x^2}{2} + 7x + C$

$= x^4 + x^3 + x^2 + 7x + C$

ii) $\int (x^{-3}).dx$

$\frac{x^{-3+1}}{-3+1} + C = \frac{x^{-2}}{-2} + C$

$= -\frac{1}{2x^2} + C$

Question 2

a) $y = 3x^2 - 2x$ average rate of change

use $M = \frac{y_2 - y_1}{x_2 - x_1}$

$x = 1$ $x = 2$

when $x = 1$, $y = 3(1)^2 - 2(1)$
$= 1$

$(1, 1)$
$x_1 y_1$

$M = \frac{8 - 1}{2 - 1} = \frac{7}{1} = \boxed{7}$

when $x = 2$, $y = 3(2)^2 - 2(2)$
$= 8$

$(2, 8)$
$x_2 y_2$

b) $y = 4x^3 - 12x + 2$

note: $x^0 = 1$

i) $dy/dx = 4(3)x^{3-1} - 12(1)x^{1-1} + 0$ use power rule if $y = x^n$ $dy/dx = nx^{n-1}$

$= 12x^2 - 12$

$d^2y/dx^2 = 12(2)x^{2-1} - 0$

$= 24x$

ii) slope of tangent line where $x = 0$.

$dy/dx = 12x^2 - 12$

sub in $x = 0$: $12(0)^2 - 12$

$= 0 - 12$

$= \boxed{-12}$

iii) coordinates of Max and Min turning points:

$dy/dx = 12x^2 - 12$ $\quad d^2y/dx^2 = 24x$

let $dy/dx = 0$ & solve for x

$12x^2 - 12 = 0$

$12x^2 = 12$

$x^2 = 12/12 = 1$

$x = 1$ or $x = -1$

when $x = 1$ $d^2y/dx^2 = 24(1)$ Min point

$= 24$

when $x = -1$ $d^2y/dx^2 = 24(-1)$

$= -24$ Max point

when $x = 1$ $y = 4(1)^3 - 12(1) + 2$

$= -6$

Min turning point at $(1, -6)$

when $x = -1$, $y = 4(-1)^3 - 12(-1) + 2$

$= 10$

Max turning point at $(-1, 10)$

c) $I = 1 + 20t - 5t^2$

i) current after 1 second ∴ sub 1 in for t

$I = 1 + 20(1) - 5(1)^2$

$= 16$ milliamps

ii) rate after 1.5 seconds

— get the derivative.

$\frac{dI}{dt} = 20 - 10t$ sub in $t = 1.5$

$= 20 - 10(1.5)$

$= 5$ milliamps per second.

iii) time to reach peak ∴ let $\frac{dI}{dt} = 0$ & solve for t

$\frac{dI}{dt} = 20 - 10t = 0$

$-10t = -20$

$10t = 20$

$t = \frac{20}{10}$

$t = 2$ seconds

(definite integral)

d) area enclosed by curve $y = 2x^2 + 3x + 1$ $x = 1$ $x = 3$.

$\int_1^3 (2x^2 + 3x + 1) \, dx$

$\left[\frac{2x^3}{3} + \frac{3x^2}{2} + x\right]_1^3 = \left[\frac{2(3)^3}{3} + \frac{3(3)^2}{2} + (3)\right] - \left[\frac{2(1)^3}{3} + \frac{3(1)^2}{2} + (1)\right]$

$= \boxed{31.33}$